

PRELIMINARY PHARMACOLOGICAL STUDIES ON SOME TRADITIONAL HERBAL MEDICINES

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ABSTRACT

Four traditional medicines, Tiryag-e-Fishar, Dawa-ul-Misk Motadil Sada, Durr-e-Seer and Jigreen CL have been studied for their blood pressure lowering effect and acute toxicity. Tiryag-e-Fishar and Dawa-ul-Misk showed high potency as hypotensive agents while, Durr-e-Seer was found to be experimentally inactive. Jigreen CL, which is a drug prescribed for liver diseases, was found devoid of any lethal effect at the dose of 20 ml/kg.

INTRODUCTION

Herbal drugs have been used for thousands of years in Traditional system of medicine as major remedy for treatment of many human ailments and to maintain human health (Saeed, M. 1969, Chan K.L., 1993, Dymock W., 1890). They are still extremely effective and now becoming popular as alternative medicines especially in developing countries, where people can not afford expensive foreign drugs. However, only a few have been studied using the modern sensitive techniques to ascertain their pharmacological and toxicological properties. It is therefore required to produce enough evidence by common scientific approaches to examine the safety and efficacy of herbal medicine.

Keeping these factors in view, a program was initiated by which efficiency of these drugs could be examined on animals in terms of their activity and toxicity. The pharmacological activities were determined on modern sophisticated instruments while toxicity test followed the recognized protocol (Fatima, N., 1993). Present paper deals with the hypotensive activity and toxicity test of four traditional medicines, Durr-e-Seer, Tiryag-e-Fishar, Dawa-ul-Misk and Jigreen CL.

MATERIALS AND METHODS

Preparation of Sample:

Durr-e-Seer tablets were crushed into a fine powder by pestle and mortar.

Powder was dissolved in saline and centrifuged at 4000 revolutions per minute (rpm) for 15 minutes. Supernatant, containing approximately 50% of the drug was subjected to preliminary pharmacological screening. Tiryaq-e-Fishar and Dawa-ul-Misk Motadil Sada when dissolved in saline, 20% of the drug settled down and remaining upper part made easily injectable suspension.

Hypotensive Activity:

Hypotensive effect of Durr-e-Seer, Tiryaq-e-Fishar and Dawa-ul-Misk were studied in normotensive Sprague Dawley rats (200-250g), anaesthetized with pentobarbital sodium (60 mg/kg i.p.). The blood pressure was recorded from the carotid artery via arterial cannula connected to a blood pressure transducer (60-3003, Harvard) coupled with four channel Harvard Universal Oscillograph (50-9307). Drugs were injected through a cannula inserted in jugular vein. Blood pressure of treated animals was expressed as percent decline in the blood pressure with respect to control values (Dar, A., 1997). Acetylcholine (1 μ g/kg) was used as a positive control which produced $57.67 \pm 1.3\%$ (Mean $\pm$ SEM, $n=3$) fall in the blood pressure.

Toxicity Test:

Acute toxicity was studied on NMRI-1 mice (20-25 g). Two sets of 10 mice (5 animals per sex) were given 20 ml/kg of Jigreen CL orally and regularly at the interval of 24 hrs. Animals were observed for three days.

RESULTS AND DISCUSSION

Preliminary studies on hypotensive activity of three traditional drugs manufactured in Hamdard Laboratories were carried out on Sprague Dawley rats. Drugs examined were Durr-e-Seer, Tiryaq-e-Fishar and Dawa-ul-Misk Motadil Sada. Durr-e-Seer, is a dry powder of *Allium sativum* which is commonly known as garlic in English and lahsan in Urdu and Hindi. It is considered to be a tonic and regarded as useful in heart disease, sciatica, leprosy, asthma and rheumatism. Its juice is applied externally as a counter irritant (Dymock, W., 1890). In the form of tablets, it is useful for general debility of nerves, cough, bronchitis, high blood pressure and abdominal disorders (Hamdard). Drug was soluble approximately 50% in saline at neutral, acidic (pH 4), alkaline pH (pH = 11) and in 5% dimethyl sulfoxide. Soluble portion of drug in each of the above mentioned solvents did not show any change in arterial blood pressure at the dose of 50 mg/kg. As Durr-e-Seer is considered to be an antihypertensive drug (recommended human dose is 10 mg/kg, four times a day), therefore, it may be assumed that the active factor of drug may reside in its water insoluble part.

Tiryaq-e-Fishar, as indicated by name is prescribed as an antihypertensive drug

(Saeed, M., 1969). Its composition is given in Table-1 [Hamdard Laboratories (Waqf) Pakistan]. Some properties of its ingredients have also been mentioned in Table-1 (Nadkarni, K.M., 1976; Wealth of India, 1948-1976; Chopra N.R. 1958). 80% of drug forms an easily injectable suspension with saline. This water suspended part of drug showed 35% fall in arterial blood pressure at the dose of 80 mg/kg for 10 sec. Second application of the same dose resulted in 70% hypotension and the effect lasted for 1.5 min. Third administration of this dose caused an immediate fall in blood pressure and sudden death of the animal (Fig.1). The behaviour of drug is clearly indicative of the fact that accumulation of drug may have caused the death of rat. It would be interesting to note that recommended dose of Tiryag-e-Fishar for human being is 85 mg/kg once a day. Decrease in blood pressure of animals may be due to the sedative and diuretic properties of the most of the plants used in preparation of Tiryag-e-Fishar. Out of six animals tested, three did not respond to the drug at this particular dose.



Fig.1: Hypotensive effect of Tiryag-e-Fishar in mice at the dose of 80 mg/kg.

Dawa-ul-Misk Motadil Sada is an effective medicine for hepatic dysfunctions, anaemia and sluggish liver (Saeed, M., 1969). Herbal composition of the drug is given in Table-2 [Usmanghani, K., 1997; Hamdard Laboratories (Waqf) Pakistan]. It is also regarded as a cardiac stimulant and as such it shows corresponding rise in blood pressure, however, our studies have shown hypotensive activity in normotensive rats. Intravenous administration of hydrophilic portion of drug (which like Tiryag-e-Fishar was 80%) caused 18% and 50% fall in systolic and diastolic blood pressure at the doses of 40 mg/kg and 80mg/kg respectively (Table-3) (recommended human dose is 114 mg/kg, once a day). It is worthy to note that in case of Dawa-ul-Misk Motadil

Sada, decline in blood pressure was not immediate but took 10-30 seconds to produce effect (Fig. 2). It indicates that this medicine may be a prodrug which converts into hypotensive drug or component. Like Tiryaq-e-Fishar, potency of effect was not constant in Dawa-ul- Misk Motadil Sada. Some rats failed to give any response at the dose of 80 mg/kg, while in some cases second application of dose (80 mg/kg) led to the expiry of the animal.



Fig.2 : Hypotensive effect of Dawa-ul-Misk Motadil Sada in mice at the dose of 80 mg/kg.

Jigreen CL is used in Eastern system of medicine for enlarged and sluggish liver, ineffective hepatitis and all its inflammatory conditions (Hamdard). *Achillea millefolium*, *Artemisia absinthium*, *Tamarix dioica*, *Melia azadirachta*, *Rosa damascena*, *Sphaeranthus indicus*, *Rheum emodi* and *Allium sativum* are the herbal components of drug. The first two plants are well known for the treatment of sluggish liver. Jigreen CL was found devoid of any fatality when subjected to toxicity test and no physical symptoms were found in 72 hours at the dose of 20 ml/kg in mice (recommended human dose is 10 ml/kg, thrice a day).

Table 1
Composition of Tiryag-e-Fishar

Scientific Name	Common Name	Biological Activity
<i>Cypraea moneta</i>	Cowry	Antianaemic, Anticatarrhal and use in asthma and cough
<i>Santalum album</i>	White sandal wood tree	Sedative, Diuretic and Antipyretic
<i>Bambusa arundinacea</i>	Tabasheer	Tonic, Sedative and Diuretic
<i>Portulaca oleracea</i>	Khurfa	Astringent and Diuretic
<i>Coriandrum sativum</i>	Coriander	Carminative, Tonic and Diuretic
<i>Rosa damascena</i>	Rose	Antipyretic and Cardiotonic
<i>Onosma bracteatum</i>	Echium	Diuretic, Tonic and Antipyretic
<i>Lagenaria vulgaris</i>	Kaddu	Diuretic and Nutritive
<i>Vateria indica</i>	Kahruba	Cardiotonic and use in chronic rheumatic inflammation of joints
<i>Mytilus margaritiferus</i>	Pearls	Sedative, Tonic and Nutritive
<i>Malus sylvestris</i>	Apple	Diuretic, Nutritive and useful in anaemia, liver and heart disorder
<i>Ambra grasea</i>	Ambergris	Fragrant, antiseptic and antispasmodic
<i>Arum</i>	Gold	Diuretic, aphrodisiac and Tonic
<i>Argentum</i>	Silver	Sedative and Tonic

Table 2
Composition of Dawa-ul-Misk Motadil Sada

Scientific Name -	Common Name	Biological Activity
<i>Bambusa arundinacea</i>	Tabasheer	Diuretic, Sedative, Antipyretic and Tonic
<i>Pterocarpus santalinus</i>	Red Sandal wood	Colouring agent, Tonic and Diaphoretic
<i>Santalum album</i>	White sandal wood tree	Diuretic, Sedative and Antipyretic
<i>Salvia haematodes</i>	Behman surkh	Cardiatonic and Refrigerant
<i>Pistacia lentisus</i>	Mastiche tree	Diuretic and Carminative
<i>Parmelia perlata</i>	Stone flowers	Diuretic, Astringent and Antidiarrhoeal
<i>Embllica officinalis</i>	Indian gooseberry	Diuretic, Refrigerant and Laxative
<i>Elettaria cardamomum</i>	Cardamom	Diuretic, Stomachic and Carminative
<i>Cinnamon officinalis</i>	--	Carminative, Astringent and Antispasmodic
<i>Coriandrum sativum</i>	Coriander	Diuretic, Carminative, Tonic and Stimulant
<i>Onosma bracteatum</i>	Ratanjot	Colouring, Astringent and Lithontriptic
<i>Berberis aristata</i>	Indian barberry	Tonic, Purgative, Stomachic and Astringent

Table 3
Hypotensive activity of aqueous phase of traditional medicines

Drugs	Dose (mg/kg)	% Fall in Arterial blood pressure (mm Hg) with $\pm$ SEM
Durr-e-Seer	50	Nil'
Tiryag-e-Fishar	40	Nil
	80	52.50 $\pm$ 12.64
Dawa-ul-Misk Motadil	40	18.11 $\pm$ 2.67
Sada	80	53.37 $\pm$ 1.52

Nil showed no significant fall in blood pressure.

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